

Griffith School of Engineering and Built Environment  
Griffith University

## 3004ENG — Project Management Principles

Trimester 2, 2026

### Eris TestFlight2 Launch Campaign

#### Project Execution Plan

Assessment 3: Group Project Execution Plan — Group 1

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Acting as a project management consultancy engaged to deliver launch-site operations at Bowen Orbital Spaceport.

## Contribution

Table 1: Agreed contribution after hours equalisation, each member 42 h.

| Team member                  | Hours      | Primary deliverables                                                 | %            |
|------------------------------|------------|----------------------------------------------------------------------|--------------|
| Abdul Wahab Obeid (s5303839) | 42         | Executive summary; integration; governance; delivery model; final QA | 25.0         |
| Hadi Alkhub (s5331680)       | 42         | Scope; schedule; network logic; Gantt; compression                   | 25.0         |
| Ziyad Alahmari (s5375826)    | 42         | Risk register; quality and HSE                                       | 25.0         |
| Ilsa Hashmi (s5310047)       | 42         | Cost; resources; capacity; EVM framework                             | 25.0         |
| <b>Total</b>                 | <b>168</b> |                                                                      | <b>100.0</b> |

We confirm that the contribution percentages above and the submission below are agreed by all members.

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# 1 Executive summary

This Project Execution Plan asks the Gilmour Executive Board to approve or decline Eris TestFlight2 *launch-site operations* at Bowen Orbital Spaceport from this section alone. TestFlight 1 cleared the tower and ended at T+14 s after thrust loss in two first-stage hybrid motors, from electrical and thermal pump faults (Smith, 2025; Gilmour Space Technologies, 2025; Dougherty, 2026). Group 1 is a project-management consultancy. In-scope work is pad operations: cleanroom receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second static fire of the four first-stage hybrid engines, range clearance, lift-off, telemetry handover and closeout. Factory manufacture, engine redesign and freight to the receiving gate stay with Gilmour and sit outside this baseline.

The work estimate is \$1,500,000. Contingency is \$78,890 (5.26% of base): \$35,760 of residual expected monetary value on ten cost-bearing risks plus a \$43,130 itemised in-scope residual-uncertainty allowance. Authorised *BAC* is \$1,578,890. Contingency may be consumed; nothing beyond *BAC* is authorised. Delivery is a managing-contractor / design-and-construct hybrid: Gilmour keeps propulsion intellectual property, vehicle integration and the launch licence; civil and cryogenic GSE packages are let design-and-construct with an interface-control freeze and liquidated damages.

The campaign uses a seven-day pad calendar from 10 August to 1 December 2026 because LOX load, static fire and *T-0* do not pause for weekends. Stacking completes on 15 October 2026 (M-4 / A-123). Cold Wet Dress Rehearsal completes on 24 October (A-132). The five-second static fire completes on 30 October 2026 (M-5 / A-133). Launch Readiness Review is on 10 November 2026 (M-6 / A-141). Baseline *T-0* is **15 November 2026** (M-7 / A-143), inside the charter's Q1 2027 outer window. Telemetry handover is on 20 November 2026 (M-8 / A-144), with at least 98.5% of packets. Closeout is on 1 December 2026 (M-9 / A-145). A five-day pull-in to 10 November exists only if R-14 is realised, at a contingent \$21,000 draw on the reserve. It is not the baseline.

Three register items can stop *T-0* or close the heritage case: R-01, R-12 and R-14. R-01 is a residual pump-insulation defect: static-fire auto-abort on A-133 and a 12–14 day critical-path recycle. R-12 is the TestFlight 1 in-flight mode: thrust loss in two first-stage hybrid motors after *T-0*, as at T+14 s, ending the heritage objective. R-14 is an ASA flight-authorisation or CASA NOTAM refusing the 15 November slot. Occupancy is held at the pad occupancy cap of twelve after an unmitigated peak of fourteen. The organic crew is eight.

*It is recommended that the Executive Board approve this Project Execution Plan, baseline the campaign at \$1,578,890 AUD (inclusive of \$78,890 contingency), including the \$602,301 of planned pad PV already on the clock by Gate 1 on 2 October 2026 (40.2% of the work PMB: permitting, Juru, cleanroom, GSE, RF and stacking in progress), lock remaining spend at that gate, and hold T-0 on 15 November 2026 inside the charter's Q1 2027 outer window.*

## 2 Refined project overview

### 2.1 Background and purpose

This is the Week 11 reading of a campaign chartered on 7 August 2026. Gilmour Space Technologies flew Eris TestFlight 1 from Bowen in July 2025. The vehicle cleared the tower and was terminated at T+14 s after thrust loss in two first-stage hybrid motors (Smith, 2025; Gilmour Space Technologies, 2025; Australian Space Agency, 2025). TestFlight 2 is the late-2026 follow-on, inside the charter's Q1 2027 outer window (Dougherty, 2026). Group 1 plans and controls *launch-site operations*: cleanroom receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second static fire of the four first-stage hybrid engines, range clearance, lift-off on 15 November 2026, handover of at least 98.5% of telemetry within five days, and closeout. Manufacturing, engine research, freight to the gate, and pump redesign remain Gilmour factory work. Orbital insertion is Gilmour's vehicle-success test, not a consultancy finish-line metric.

The Bowen facility licence has been held since March 2024. Stacking (A-123) is the mobile-crane vertical integration that stands the 23 m vehicle on the pad: stages are lifted, mated and bolted down, with no ignition. That is site work, so the existing licence already authorises it. Hot fire is the five-second static fire of the four first-stage hybrid engines (A-133): first ignition on the pad, vehicle still hold-down, not lift-off. That is a range activity, so it still waits on Australian Space Agency (ASA) flight authorisation, Civil Aviation Safety Authority (CASA) airspace, and Great Barrier Reef Marine Park Authority (GBRMPA) marine-park permissions (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019; Australian Space Agency, 2025). Those three instruments therefore gate hot fire, not stacking. ASA stays Keep Satisfied, with filings 60 days prior, as the charter scored it; that filing sits on A-113. CASA issues the Notice to Airmen (NOTAM) as the instrument of A1 M-3 civil aviation airspace arrangements; it does not replace ASA. The Juru people stay High power / High interest, Manage Closely. That class is unchanged. The new control is monitors with stop-work authority on the network before Wet Dress Rehearsal. The payload sponsor stays Keep Informed and witnesses LRR on A-141; payload operations remain out of scope and have no WBS package.

### 2.2 Scope

Scope is on the site operations. Included: campaign governance and EVM (WBS 1.1), pad logistics and GSE (1.2), pad testing (1.3), and flight, launch and closeout (1.4). Figure 2.1 has seventeen Level-3 packages; those are the cost accounts. Table 3.1 has nineteen schedule bars. The extra two are the split of package 1.2.1 into cleanroom (A-121) and unpacking (A-122), and the external vehicle-receipt wait (A-100). Excluded: factory manufacture, engine R&D, line-haul to the gate, payload operations, and any unidentified Level-3 box that would mix factory

product into a site baseline. A-100 is a zero-dollar external driver that pins A-122; it is not a consultancy-controlled launch-critical bar. It is omitted from Table 4.1. Design defects found on the pad are packaged back to R&D. Only a pre-kitted spare may be swapped on A-134.

## 2.3 Charter baseline versus PEP refinement

This is not a copy of the Week 4 charter. Table 2.1 lists every main change and why it was made. Two Assessment 1 dates did not match: stacking complete (M-4) is now A-123 on 15 October 2026, and telemetry handover (M-8) is now A-144 on 20 November.

Short forms used in the table: LRR (Launch Readiness Review), LEO (Low Earth Orbit), HSE (Health, Safety and Environment), EMV (expected monetary value), RES (residual-uncertainty allowance), WDR (Wet Dress Rehearsal), and LOX (liquid oxygen).

Table 2.1: Charter baseline versus PEP refinement.

| Project element                 | A1 charter (Week 4)                                                        | A3 PEP (Week 11)                                                                                                                                           | Why it changed                                                                                               |
|---------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>M-4 stacking date</b>        | Objective 1: pad integration 15 Oct. Milestone M-4: stacking 28 Oct.       | <b>M-4 locked to 15 Oct 2026</b> (A-123 <i>EF</i> ).                                                                                                       | Removes the internal collision. Creates a 15-day pad-test buffer before the 30 Oct hot-fire gate.            |
| <b>M-8 handover date</b>        | “5 days post-launch” and milestone date 22 Nov (7 days after 15 Nov).      | <b>M-8 = 20 Nov 2026</b> (A-144).                                                                                                                          | Aligns the milestone with Objective 5 and the 5-day rule.                                                    |
| <b>Cost baseline</b>            | “Manage within the allocated operational budget.” No numbers figure.       | Bottom-up <b>\$1,500,000 + \$78,890</b> contingency (residual EMV \$35,760 + RES \$43,130).                                                                | Crane hire, cryogenic crews, and LOX delivery cannot be controlled without a numbered baseline.              |
| <b>Permit vs stacking logic</b> | M-3 flight permits 25 Oct were predecessors of M-4 stacking (then 28 Oct). | Facility licence already held (Mar 2024). <b>A-113 gates A-133 static fire, not stacking.</b> Permits <i>EF</i> 24 Oct; hot fire <i>ES</i> 25 Oct.         | Stacking is a site-licence activity. Hot fire is a range activity. This also creates a second critical path. |
| <b>Juru engagement</b>          | High power / high interest, Manage Closely, consultation and monitoring.   | Same power/interest (High / High, Manage Closely). <b>Embedded monitors with stop-work authority</b> and a cultural hold point before WDR (A-114 → A-132). | A1 strategy was correct in class; the PEP makes it an enforceable pad control, not a comms plan.             |
| <b>Testing sequence</b>         | WDR and 5-second static fire as one Objective 3 output by 30 Oct.          | <b>A-132 WDR (cold) 16 Oct–24 Oct, then A-133 static fire 25 Oct–30 Oct.</b> LRR waits on A-134 diagnostics.                                               | Hot fire must wait for flight permits. LRR cannot start on incomplete post-fire data.                        |

Table 2.1: Charter baseline versus PEP refinement (continued).

| Project element              | A1 charter (Week 4)                                                                                           | A3 PEP (Week 11)                                                                                                                                                                                                                                                         | Why it changed                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk governance</b>       | Five high-level threats, no $P \times I$ money.                                                               | <b>18 campaign-specific risks</b> , Cause / Event / Effect columns, EMV \$35,760 + RES \$43,130.                                                                                                                                                                         | Generic rows cannot size contingency or point at a WBS line.                                                                             |
| <b>Resource peak</b>         | Roles listed; no demand vs capacity.                                                                          | Organic crew <b>8</b> named heads (GSE1, GSE2, GSE3, GSE4, STK1, STK2, LD, RSO), pad HSE cap <b>12</b> , unmitigated peak <b>14</b> , mitigated peak <b>12</b> .                                                                                                         | A1 could not show where the plan exceeds capacity.                                                                                       |
| <b>Regulator and payload</b> | ASA High/Low Keep Satisfied, file 60 days prior. Payload sponsor Med/High Keep Informed, witness LRR.         | ASA remains Keep Satisfied and is filed 60 days prior on A-113. CASA issues the NOTAM as the instrument of A1 M-3 civil aviation airspace arrangements, not as a substitute for ASA. Payload sponsor still witnesses LRR on A-141 (HP-4). No 10–12 Nov geometry is used. | A1 named ASA, not Townsville, as the high-power regulator. The PEP keeps that filing rule and uses CASA only as the airspace instrument. |
| <b>Success criteria</b>      | Q1 2027 window; no funds beyond contingency; 98.5% telemetry in 5 days; zero LTI/GBRMPA/Juru; defects to R&D. | T-0 15 Nov inside A1's Q1 2027 window (2-day in-flight trigger); $AC \leq BAC$ ; 98.5% telemetry in 5 days; LRR first submission; four-engine 5 s fire; zero LTI/GBRMPA/Juru; defects to R&D. LEO is Gilmour's test, not the consultancy's.                              | A1 tests stay. The PEP numbers cost and tightens in-flight control inside Q1 2027. LEO is not this team's finish line.                   |
| <b>Objectives 4 and 5</b>    | Objective 4: pass LRR by 10 Nov. Objective 5: November 2026 window and 98.5% telemetry.                       | Objective 4 is LRR by 10 Nov (M-6 / HP-4, first submission). Objective 5 is the November window (T-0 15 Nov) plus 98.5% handover on 20 Nov. Closeout on 1 Dec is a success test, not a numbered objective.                                                               | A3 keeps A1's numbered (Obj 4 as LRR rather than remapping it onto T-0.                                                                  |
| <b>LRR attendees</b>         | M-6: Launch Director, Range Safety Officer and Juru Land Liaisons review. Payload sponsor witnesses LRR.      | HP-4 release is Launch Director, RSO, Juru Officer, Abdul and Ziyad. Payload sponsor remains a witness, not a signatory.                                                                                                                                                 | A1 named the review board. The PEP keeps those three on the hold and does not promote the payload sponsor from witness to signatory.     |

Table 2.1: Charter baseline versus PEP refinement (continued).

| Project element                         | A1 charter (Week 4)                                                                                                          | A3 PEP (Week 11)                                                                                                                        | Why it changed                                                                                                                            |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Community and media stakeholders</b> | Bowen Community Keep Informed: clearance maps 14 days prior. QLD Regional Press Monitor: post-launch press releases.         | Community notice 14 days prior (R-17). No named media stakeholder. Clearance maps are not a PEP artefact.                               | The 14-day community notice A1 required stays as an HSE control. A Monitor-only press party is not a pad control.                         |
| <b>WDR T-10 acceptance</b>              | M-5 acceptance: countdown to 10 seconds before simulated lift-off, plus a 5-second static fire with zero critical anomalies. | A-132 is cold WDR (16–24 Oct). A-133 is the five-second static fire (25–30 Oct). The T-10 simulated-lift-off criterion is not restated. | Cold WDR is LOX load and countdown rehearsal without ignition. T-10 simulated lift-off is not a separate success test in this PEP.        |
| <b>Ilsa Hashmi project role</b>         | Project role Stakeholder & Resource Lead. A3 column already Cost, Resource, Capacity Planning.                               | Cover: Cost, Resource & EVM Lead. Organogram: Cost and Resource Lead. Stakeholder Lead is not used.                                     | A1's A3 column already pointed at cost and resource. EVM is the A3 add. The Stakeholder Lead title is not on the PEP cover or organogram. |
| <b>Field spend before Gate 1</b>        | M-1 charter 7 Aug. M-2 / Gate 1 PEP lock 2 Oct. Phase 3 execution follows M-2.                                               | Charter has no NTP. Permitting (A-113) and Juru (A-114) start 10 Aug. Pad occupancy before Gate 1 is a PEP choice. Gate 1 PV \$602,301. | Permit and heritage clocks cannot wait until 2 Oct and still finish before hot fire. Remaining spend still locks at Gate 1.               |

## 2.4 Objectives and success criteria

The Week 4 objectives still stand. Building the network and the budget taught us to lock the dates and the dollars. Each test is checked in Sections 3–7.

Objective 1 stayed stacking. A1 also put M-4 on 28 October, so the same job had two dates and no room for cold flow before hot fire. We locked M-4 to A-123 on 15 October 2026 so the objective and the milestone match.

Objective 2 was only “manage within the allocated budget.” Crane hire, cryogenic crews and LOX cannot be run with no figure. The PEP now uses a \$1,500,000 base and \$78,890 of residual-EMV-plus-RES contingency, audited at Post-Stacking (M-4), Post-Static Fire (M-5) and Post-Launch (M-7).

Objective 3 was one 30 October output: WDR and the five-second fire together. Hot fire has to wait for flight permits, so A-132 is now cold WDR (16–24 October) and A-133 is the five-second static fire of the four first-stage hybrid engines (25–30 October).

Objective 4 stayed pass LRR by 10 November (A-141 / M-6 / HP-4). We did not move that number onto lift-off. The PEP adds that it must pass on the first formal submission.

Objective 5 stayed the November window and 98.5% telemetry. A1's M-8 sat on 22 November, two days past the five-day rule. Lift-off is 15 November (A-143 / M-7) inside the Q1 2027 window, with a more-than-two-day in-flight slip trigger, and handover on 20 November (A-144 / M-8). Closeout on 1 December (A-145 / M-9) is a success test, not a new Objective 6.

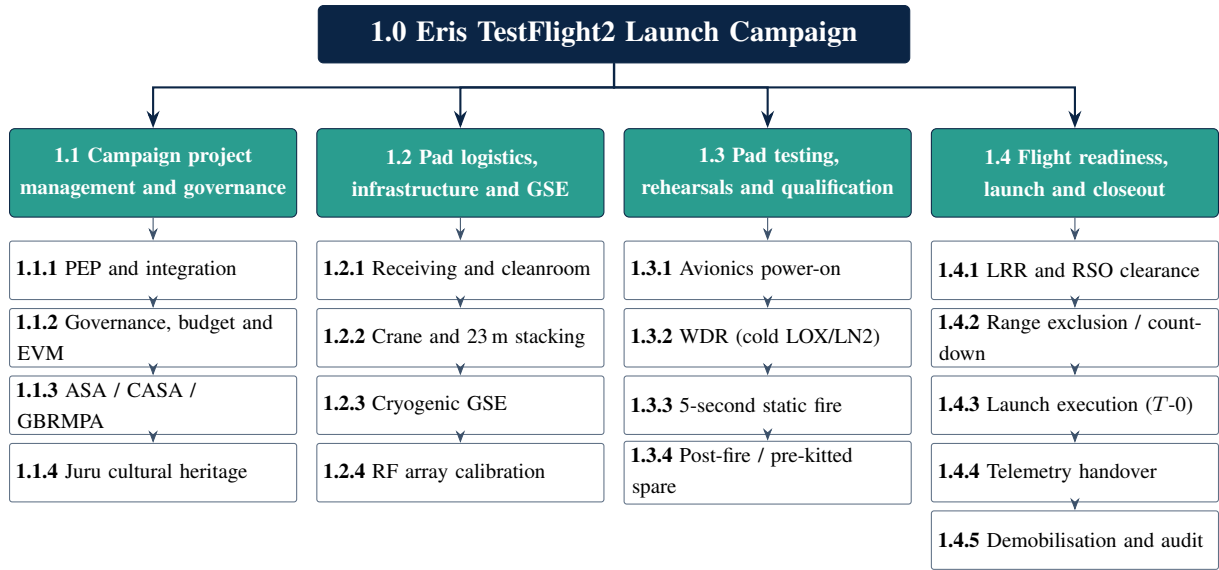
Finish-line success is the pair used in earned value: final  $AC \leq \$1,578,890$  AUD, and  $T-0$  on 15 November with that  $> 2$  day trigger. Contingency may be used; nothing above  $BAC$  is authorised. HSE stays as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches. Design defects go back to R&D. Reaching LEO is Gilmour's test, not this consultancy's.

## 2.5 Work breakdown structure

The work breakdown structure in Figure 2.1 is applied to launch-site operations at Bowen Orbital Spaceport. Four Level-2 workstreams—campaign governance, pad logistics, pad testing, and flight/closeout—decompose into seventeen Level-3 packages. Every scheduled activity except A-100 maps to exactly one of those packages, and every dollar in Table 4.1 sits in exactly one of them. A-100 is annotated beside the tree rather than inside it: the vehicle receipt window is an external zero-dollar wait that starts when out-of-scope freight reaches the gate. It belongs on the network because it drives stacking (A-122), but it is not a cost account.

Package 1.2.1 is one cost account with two schedule bars (A-121 site mobilisation and A-122 receiving inspection) because receiving cannot start until both the cleanroom and the vehicle exist. The remaining sixteen Level-3 codes are one-to-one with Table 3.1. Factory manufacture, engine research and development, and line-haul to the receiving gate are excluded.

The 7 August charter does not contain a notice to proceed for field work. Permitting (A-113) and Juru mobilisation (A-114) start on 10 August because those clocks have to run before hot fire. Physical pad occupancy before Gate 1 (A-111, 2 October) is a PEP choice, not a charter clause. What Gate 1 locks is the cost and schedule baseline.



**EXT A-100 — not a WBS cost account.** External \$0 freight driver into A-122; omitted from Table 4.1.

Figure 2.1: Work breakdown structure for Eris TestFlight2 launch-site operations. Seventeen Level-3 packages are the cost accounts in Table 4.1. A-100 is an external schedule bar, not a seventeenth-plus cost line.

### 3 Schedule management

The campaign schedule turns the WBS into a dated baseline for board approval. Dates come from a finish-to-start activity-on-node network. The pad calendar is seven days a week from 10 August to 1 December 2026, because LOX load, static fire and lift-off do not stop for week-ends. Approved lift-off ( $T-0$ ) stays 15 November 2026. That day still sits inside the charter's last-allowed window of Q1 2027. A five-day pull-in to 10 November exists only if R-14 is realised (ASA flight authorisation or a CASA NOTAM refuses the 15 November slot). It is not the baseline.

#### 3.1 Network logic and float analysis

Links are finish-to-start with zero lag, drawn activity-on-node. Textbook discrete-time notation writes  $EF = ES + D$  when  $EF$  is exclusive. Every date on the Bowen calendar is inclusive. The identities used here are

$$ES_i = \max_{j \in P_i}(EF_j) + 1, \quad EF_i = ES_i + D_i - 1, \quad (1)$$

$$LF_i = \min_{k \in S_i}(LS_k) - 1, \quad LS_i = LF_i - D_i + 1, \quad (2)$$

$P_i$  and  $S_i$  are the predecessor and successor sets of activity  $i$ . Where  $P_i$  is empty,  $ES$  is the campaign start. Total float and free float are

$$TF_i = LS_i - ES_i = LF_i - EF_i, \quad FF_i = \min_{k \in S_i} (ES_k) - EF_i - 1, \quad (3)$$

If  $S_i$  is empty,  $FF_i = TF_i$ . Launch-critical means  $TF_i = 0$  on a path that sets  $T-0$  or closeout. Table 3.1 is the full pass. It matches the locked register.

The forward pass gives two  $TF = 0$  chains. They stay independent until hot fire. The vehicle path starts at the external \$0 freight wait (A-100; not a Gilmour launch-critical bar). Receiving (A-122) follows, then stacking (A-123), Wet Dress Rehearsal (A-132) and static fire (A-133). The permit path is ASA / CASA / GBRMPA permits (A-113) into static fire. RF calibration (A-125) is a finish-to-start predecessor of static fire, so the array is live before hot fire. It is not launch-critical. After static fire the tail from post-fire diagnostics (A-134) through closeout (A-145) is one critical chain. The PEP lock (A-111) finishes on 2 October by an imposed Must-Finish-On (Gate 1). It is not launch-critical. The Week 4 charter approved the document, not field mobilisation.

The paths meet in two staged merges. Wet Dress Rehearsal waits on stacking, cryogenic GSE (A-124), avionics power-on (A-131) and Juru monitors (A-114). That is the technical merge. Static fire then waits on the rehearsal, flight permits *and* RF calibration. That is the regulatory and range-instrument merge. Launch Readiness Review (LRR, A-141) waits on post-fire diagnostics, not on static fire. Signing that flight-clearance gate on incomplete hot-fire data would break the hold-point logic of an AS9100D system (SAE International, 2016).

Feeder float is not interchangeable. Avionics has one day into the rehearsal. Cryogenic GSE has three. RF calibration has 39 days. It sits in August–September so it does not sit on the October pad peak. Moving it does not change 15 November. Cleanroom mobilisation (A-121) has  $TF = 3$  but  $FF = 0$ , so delay is absorbed before  $T-0$  only by consuming the GSE float. Total float protects  $T-0$ . Free float protects the successor. Figure 3.1 draws the links.

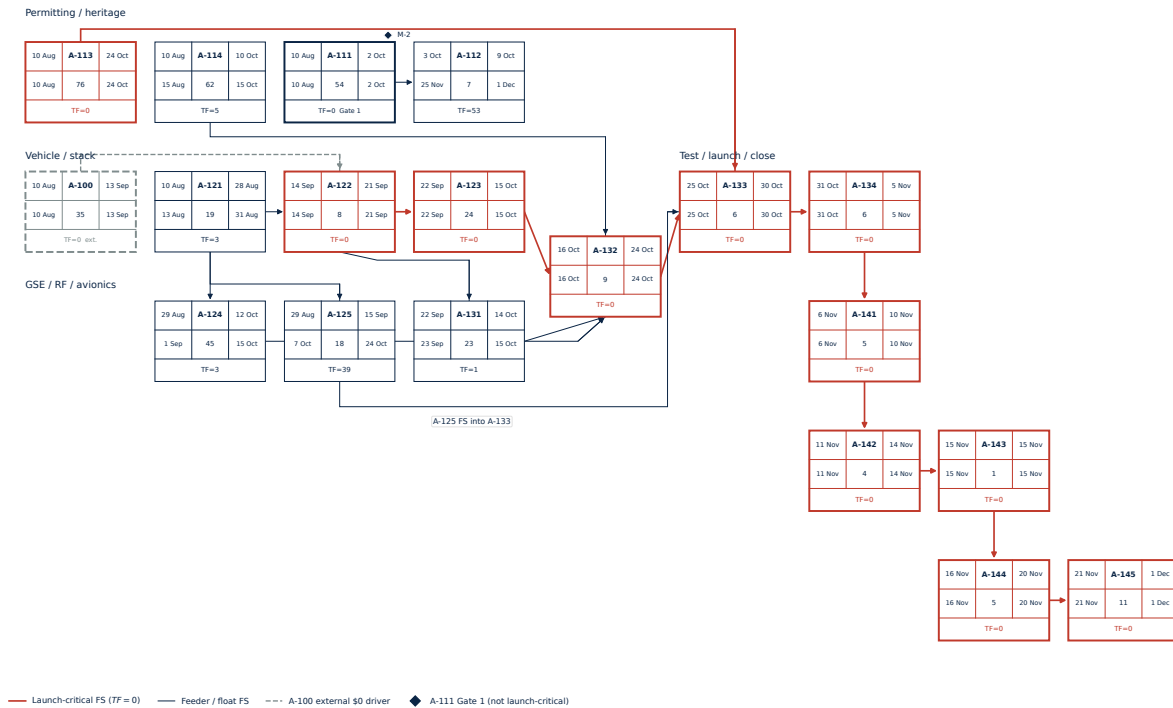


Figure 3.1: Activity-on-node network for the Bowen campaign. Node bands are  $ES / ID / EF$ ,  $LS / duration / LF$ , and  $TF$ . Crimson arrows are launch-critical finish-to-start links. The vehicle path and the permit path merge at static fire; RF calibration is the labelled south-routed feeder into the same merge; LRR waits on post-fire diagnostics, not on static fire. The PEP lock is Gate-1 constrained and is not launch-critical. Freight receipt is a grey dashed external \$0 driver, not a crimson Gilmour bar.

Table 3.1: Master campaign activity precedence and float register. Dates are inclusive.  $FF = \min(ES_{\text{succ}}) - EF - 1$  days, or  $FF = TF$  if there is no successor. The PEP lock  $TF = 0$  is an imposed Gate 1 finish, not launch-critical. The freight wait  $TF = 0$  is an external pin (CP = Ext), not a Gilmour launch-critical bar.

| ID    | WBS    | Activity                          | $D$ | Pred.                      | ES     | EF     | LS     | LF     | TF | FF | CP   |
|-------|--------|-----------------------------------|-----|----------------------------|--------|--------|--------|--------|----|----|------|
| A-100 | EXT    | Vehicle receipt (external driver) | 35  | —                          | 10 Aug | 13 Sep | 10 Aug | 13 Sep | 0  | 0  | Ext  |
| A-111 | 1.1.1  | PEP and integration baseline      | 54  | —                          | 10 Aug | 2 Oct  | 10 Aug | 2 Oct  | 0* | 0  | Gate |
| A-112 | 1.1.2  | EVM control stand-up              | 7   | A-111                      | 3 Oct  | 9 Oct  | 25 Nov | 1 Dec  | 53 | 53 | N    |
| A-113 | 1.1.3  | ASA / CASA / GBRMPA permits       | 76  | —                          | 10 Aug | 24 Oct | 10 Aug | 24 Oct | 0  | 0  | Y    |
| A-114 | 1.1.4  | Juru monitor mobilisation         | 62  | —                          | 10 Aug | 10 Oct | 15 Aug | 15 Oct | 5  | 5  | N    |
| A-121 | 1.2.1a | Site mobilisation and cleanroom   | 19  | —                          | 10 Aug | 28 Aug | 13 Aug | 31 Aug | 3  | 0  | N    |
| A-122 | 1.2.1b | Receiving and unpacking           | 8   | A-100, A-121               | 14 Sep | 21 Sep | 14 Sep | 21 Sep | 0  | 0  | Y    |
| A-123 | 1.2.2  | Crane and vertical stacking       | 24  | A-122                      | 22 Sep | 15 Oct | 22 Sep | 15 Oct | 0  | 0  | Y    |
| A-124 | 1.2.3  | Cryo GSE piping and checkouts     | 45  | A-121                      | 29 Aug | 12 Oct | 1 Sep  | 15 Oct | 3  | 3  | N    |
| A-125 | 1.2.4  | RF telemetry calibration          | 18  | A-121                      | 29 Aug | 15 Sep | 7 Oct  | 24 Oct | 39 | 39 | N    |
| A-131 | 1.3.1  | Avionics power-on sweeps          | 23  | A-122                      | 22 Sep | 14 Oct | 23 Sep | 15 Oct | 1  | 1  | N    |
| A-132 | 1.3.2  | Wet Dress Rehearsal (cold)        | 9   | A-114, A-123, A-124, A-131 | 16 Oct | 24 Oct | 16 Oct | 24 Oct | 0  | 0  | Y    |
| A-133 | 1.3.3  | 5-second static fire              | 6   | A-132, A-113, A-125        | 25 Oct | 30 Oct | 25 Oct | 30 Oct | 0  | 0  | Y    |
| A-134 | 1.3.4  | Post-fire / pre-kitted spare      | 6   | A-133                      | 31 Oct | 5 Nov  | 31 Oct | 5 Nov  | 0  | 0  | Y    |
| A-141 | 1.4.1  | LRR and RSO clearance             | 5   | A-134                      | 6 Nov  | 10 Nov | 6 Nov  | 10 Nov | 0  | 0  | Y    |
| A-142 | 1.4.2  | Range exclusion and countdown     | 4   | A-141                      | 11 Nov | 14 Nov | 11 Nov | 14 Nov | 0  | 0  | Y    |
| A-143 | 1.4.3  | Launch execution ( $T-0$ )        | 1   | A-142                      | 15 Nov | 15 Nov | 15 Nov | 15 Nov | 0  | 0  | Y    |
| A-144 | 1.4.4  | Telemetry decrypt and handover    | 5   | A-143                      | 16 Nov | 20 Nov | 16 Nov | 20 Nov | 0  | 0  | Y    |

Table 3.1: Master campaign activity precedence and float register (continued).

| ID    | WBS   | Activity                    | <i>D</i> | Pred. | ES     | EF    | LS     | LF    | TF | FF | CP |
|-------|-------|-----------------------------|----------|-------|--------|-------|--------|-------|----|----|----|
| A-145 | 1.4.5 | Demobilisation and closeout | 11       | A-144 | 21 Nov | 1 Dec | 21 Nov | 1 Dec | 0  | 0  | Y  |

### 3.2 Schedule baseline, Gantt visualisation and milestone integrity

The dates in Table 3.1 *are* the schedule baseline. Figure 3.2 plots the same  $ES/EF$  values. Launch-critical bars are crimson. Remaining total float is a teal whisker to  $LF$ . Permitting, heritage, mobilisation and GSE run through August and September. October is the physical peak: stacking 15 October, WDR 16–24 October, static fire 25–30 October. November is the flight-clearance tail.

Two Week 4 collisions are closed here. The charter dated pad integration 15 October in Objective 1 and stacking as M-4 on 28 October. The PEP locks M-4 to stacking on **15 October 2026**. That restores the cold-flow window before hot fire. Handover “within five days of lift-off” was also dated M-8 at 22 November, seven days after a 15 November  $T-0$ . M-8 is now telemetry handover on **20 November**. The facility licence is already held, so permits gate static fire, not stacking. Treating flight permits as a stacking predecessor was an initiation error.

Three gates sit on the baseline. Gate 1 is the PEP lock on 2 October. Gate 2 is permits on 24 October. Without those instruments static fire cannot start. That path is already  $TF = 0$ . Gate 3 is LRR on 10 November. Crashing stacking without crashing the permit path therefore cannot pull  $T-0$ .

### 3.3 Schedule compression: crashing and fast-tracking

Fast-tracking overlaps sequential work, usually at zero direct cost. Crashing adds resource on critical work and is paid as a cost slope. The Board question is whether  $T-0$  can move from 15 November to 10 November if R-14 is realised (ASA / CASA refuses the 15 November slot). Five days must come off a  $TF = 0$  path that actually sets lift-off. After the static-fire merge that path is the tail from diagnostics through lift-off. Crashing stacking by five days saves **zero** days on  $T-0$  while permits still finish on 24 October. Overlapping stacking with cryogenic GSE also saves zero days. GSE already runs in parallel with stacking.

The crash-cost slope on a crashed activity is

$$\text{Cost slope} = \frac{\Delta C}{\Delta T} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal duration} - \text{crash duration}}. \quad (4)$$

Table 3.2 and Figure 3.3 compare three options that each claim five days. Only Option C is both CPM-valid and acceptable against the LRR hold.

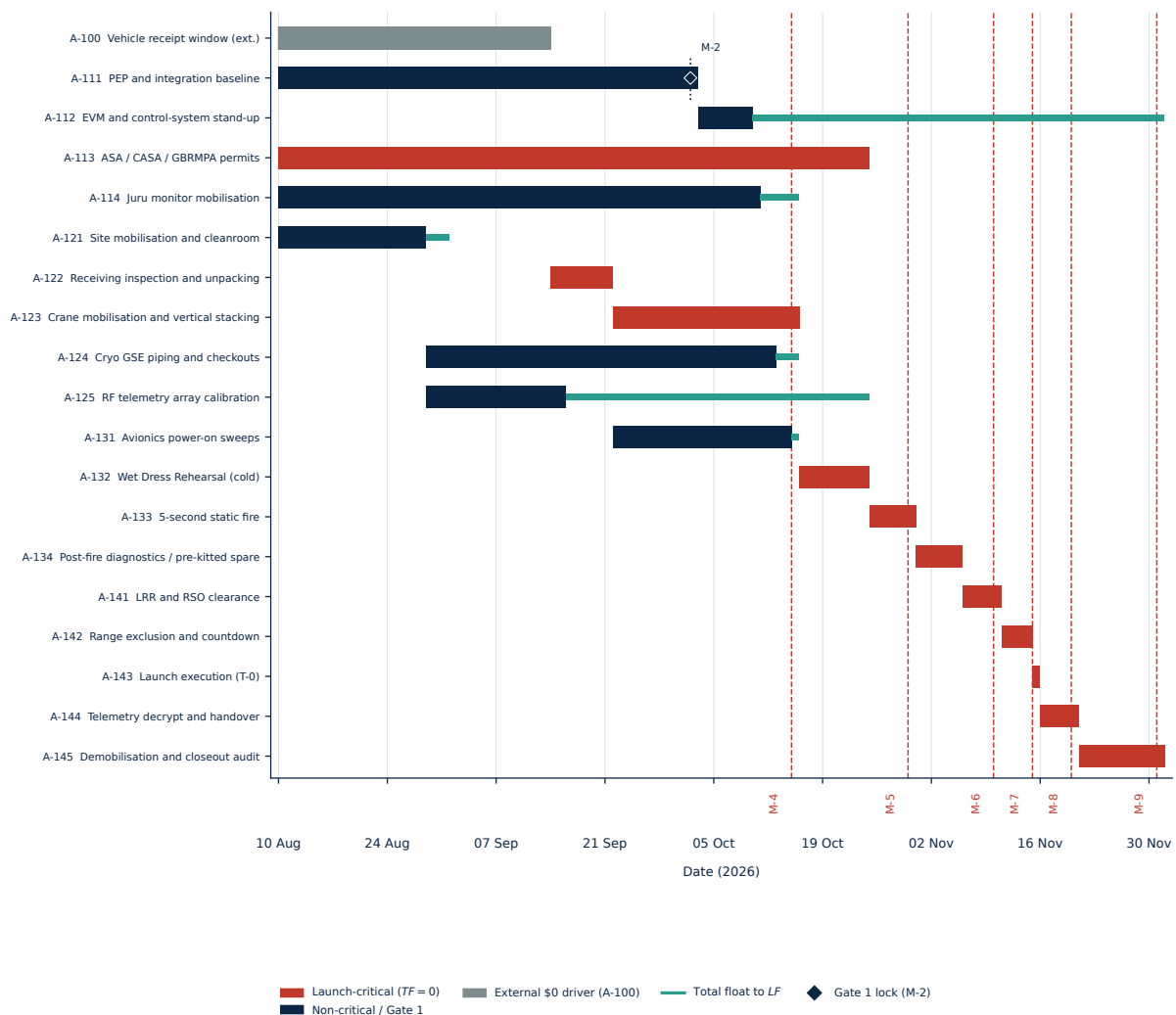


Figure 3.2: Baseline Gantt from 10 August to 1 December 2026, plotted from the same  $ES/EF$  values as Table 3.1. Crimson bars are launch-critical. The hatched grey bar is the external \$0 freight wait. Teal whiskers show total float to  $LF$ . Vertical markers are M-2 (navy, Gate 1) and M-4 through M-9. Option C is not drawn on this baseline.

Table 3.2: Schedule compression evaluation. All three options save five days on  $T-0$  if they work as drawn. Baseline  $T-0$  stays 15 November. Option C is a draw on the \$78,890 reserve if R-14 fires, not an addition to the \$1,500,000 base.

| Option                   | Activities modified                                                                 | Days | Added cost      | Slope            | Secondary risks                                                            | Verdict    |
|--------------------------|-------------------------------------------------------------------------------------|------|-----------------|------------------|----------------------------------------------------------------------------|------------|
| A Pure fast-track        | Diagnostics → LRR SS+1 (review overlaps diagnostics)                                | 5    | \$0             | \$0/d            | Review before diagnostic close-out; LRU rework; AS9100D hold-point failure | Rejected   |
| B Pure crash             | Diagnostics crashed 6 → 1 d                                                         | 5    | \$40,000        | \$8,000/d        | Fatigue; skipped thermal/electrical screens; recycle into LRR              | Rejected   |
| C Hybrid (if R-14 fires) | Diagnostics 6 → 1 d extra labour (\$21,000); LRR kept at 5 d; countdown kept at 4 d | 5    | <b>\$21,000</b> | <b>\$4,200/d</b> | Screens kept; R-15 4 d marine hold intact                                  | Contingent |

Decision: baseline  $T-0$  stays 15 November. If R-14 fires, the Board draws Option C and lift-off moves to 10 November. That is the only option that is valid on the network and acceptable on quality.

Option A is a pure fast-track. It buys five days at \$0 by starting the review one day after diagnostics start. The review board would sit while those diagnostics were still open. AS9100D is the quality rule this campaign uses for hold points (SAE International, 2016). The LRR hold (HP-4) is sign-off of that review. A hold point is a stop: the next job cannot start until the planned checks are done and the hold is released. Option A would let flight clearance start before post-fire data are complete. Signing that gate on incomplete data is a hold-point failure. Waiting to sign until diagnostics finish does not save five days; it creates rework. Fast-tracking the review is rejected.

Option B is a pure crash. It compresses diagnostics from six days to one. The slope is  $\$40,000/5 = \$8,000$  per day. A 24-hour diagnostic war room on a vehicle that has just fired is the work most likely to skip thermal and electrical screens. Crashing diagnostics without extra screens is rejected.

Option C is the hybrid. It takes the five days only from diagnostics. Extra labour keeps the thermal and electrical screens that Option B would skip: two specialists and two technicians for five days ( $2 \times \$165 + 2 \times \$95 = \$520/\text{h} \times 8 \text{ h} = \$4,160/\text{d}$ , published as the same  $\$4,200/\text{d}$  unit used on the crawler) for  $\$21,000$ . The slope is  $\$21,000/5 = \$4,200$  per day. Launch Readiness Review stays five days. Countdown stays four days, so the marine hold on R-15 is not consumed. If invoked, diagnostics sit on 31 October only, the review on 1–5 November, countdown on 6–9 November, and lift-off on **10 November**. The  $\$21,000$  is drawn from the  $\$78,890$  contingency only if R-14 is realised. It is not loaded into the  $\$1,500,000$  base. Figure 3.2 is not redrawn. Feeder floats do not tighten: avionics stays at  $TF = 1$ , cryogenic GSE at  $TF = 3$ .

Option C is chosen because it is cheaper than Option B ( $\$4,200/\text{d}$  not  $\$8,000/\text{d}$ ), it keeps the LRR hold that Option A would break, and crashing stacking cannot move  $T-0$  while permits still finish on 24 October.

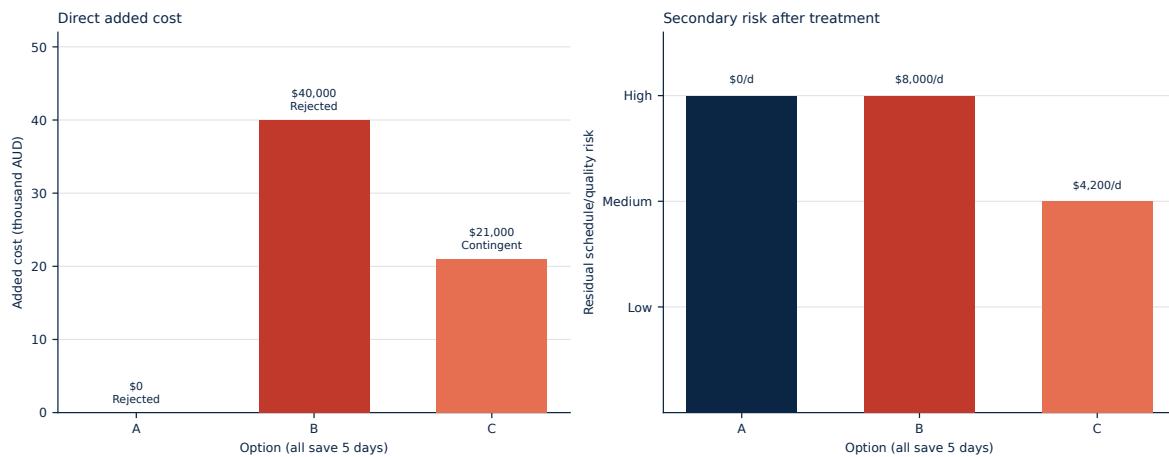


Figure 3.3: Compression trade-off for a five-day pull-in of  $T-0$ . Left: direct added cost. Right: residual schedule and quality risk after treatment. Option C is the only CPM-valid pull-in that keeps LRR at five days, countdown at four days, and pays  $\$4,200$  per day rather than  $\$8,000$ .

## 4 Cost management

The cost baseline is the WBS in Figure 2.1 priced at Level 3. Seventeen control accounts sum to a work estimate of  $\$1,500,000$ . Contingency of  $\$78,890$  is sized from residual expected monetary value plus four named residual items, not from a round-number markup, so the authorised budget at completion is  $\$1,578,890$ .

### 4.1 Estimating basis and assumptions

The estimate is bottom-up at Level 3, with analogous plant and commodity rates beside the hours. Loaded 2026 Australian-dollar rates are used throughout: systems-engineering and project-management time at  $\$150/\text{h}$ , pad technicians at  $\$95/\text{h}$ , cryogenic and range-safety specialists at  $\$165/\text{h}$ , and surge technicians at  $\$100/\text{h}$ . The Manufacturing Award C10 ordinary

hourly rate is \$29.45 from the first full pay period on or after 1 July 2026 (Fair Work Commission, 2026). Charge-out is C10 multiplied by a named employment on-cost, then utilisation and overhead—the product is in the notes to Table 4.1. Those factors do not change the \$1,500,000 lines. The 100 t crawler for 23 m stacking is priced at \$4,200 per wet day, including operator, from published Queensland plant-hire schedules cross-checked against Rawlinsons (Rawlinsons, 2025).

Inclusions are the seventeen packages in Figure 2.1. Factory manufacture, engine research and development, and freight to the receiving gate remain out of scope, so A-100 is a zero-dollar external driver rather than a cost account. The four-person surge crew (\$22,400) is inside WBS 1.3.3 as planned capacity. The \$21,000 hybrid compression cost in Section 3.3 is a contingent draw against the \$78,890 reserve if R-14 is realised; it is not in the \$1,500,000 performance measurement baseline.

Permitting and Juru work start on 10 August because those durations must finish before hot fire. Other August–September pad cost is a PEP choice. By Gate 1 (A-111, 2 October), \$602,301 of planned work is already on the clock: permits, Juru, cleanroom, GSE, RF and stacking in progress. That is 40.2% of \$1,500,000. It is planned value, not cash already paid. The leftover is \$897,699. Gate 1 still authorises the whole campaign. Locking remaining spend means those leftover package totals, plus the \$78,890 contingency rules, become the control baseline. After that, pad work cannot be added and *T*-0 cannot move without a Board variation. 2 October is not day one of spend.

## 4.2 WBS cost baseline

Table 4.1 is the rollup. Pad logistics (1.2) is \$518,000 (34.5% of the work estimate), pad testing (1.3) \$414,000 (27.6%), governance (1.1) \$300,000 (20.0%), and flight/closeout (1.4) \$268,000 (17.9%).

The largest package is crane mobilisation and stacking (1.2.2, \$218,000) because A-123 occupies 24 calendar days on the launch-critical path. The wet crawler is on hire for that whole window: 8 lift days and 16 weld, NDT and bolt-up days at \$4,200, which is \$100,800, with the balance in rigger labour. Those 16 days are plant duration, not weather. Weather dollars sit in one place only: RES four extra wet days at \$4,200 = \$16,800. R-06 is a schedule control (dawn-lift window) with residual EMV \$0, so it is not a second crane line. Wet Dress Rehearsal (1.3.2) carries \$68,500 of bulk liquid oxygen and nitrogen inside \$160,000. Static fire (1.3.3) carries the planned surge inside \$148,000. The seventeen lines sum to \$1,500,000 with no miscellaneous remainder. Package 1.2.1 remains one control account for A-121 and A-122.

## 4.3 Contingency

Contingency is not a 15% story. Table 4.2 adds a residual expected-monetary-value pool to four named residual items. Treatments already sit in the \$1,500,000 base (surge hire, dawn-lift

Table 4.1: Bottom-up cost baseline by Level-3 WBS package. Totals equal the work estimate of \$1,500,000. A-100 is omitted because it is not a cost account.

| WBS         | Package                      | Amount (AUD)     | Basis                                                          |
|-------------|------------------------------|------------------|----------------------------------------------------------------|
| 1.1.1       | PEP and integration          | 72,000           | 480 h PM @ \$150                                               |
| 1.1.2       | Governance and EVM           | 48,000           | 320 h planner/cost @ \$150                                     |
| 1.1.3       | ASA / CASA / GBRMPA          | 88,000           | ASA \$42k + CASA \$28k + GBRMPA \$18k                          |
| 1.1.4       | Juru monitors and heritage   | 92,000           | Two monitors + survey                                          |
| 1.2.1       | Receiving and cleanroom      | 68,000           | Technician hours + consumables                                 |
| 1.2.2       | Crane and 23 m stacking      | 218,000          | 8 lift + 16 weld/NDT/bolt-up d × \$4,200 = \$100,800 + riggers |
| 1.2.3       | Cryo GSE                     | 180,000          | Valves/piping + specialist crew                                |
| 1.2.4       | RF array calibration         | 52,000           | Crew + analyser hire                                           |
| 1.3.1       | Avionics power-on            | 58,000           | Engineers @ \$150/h + EGSE                                     |
| 1.3.2       | WDR (cold)                   | 160,000          | Includes bulk LOX 25 t and LN2 \$68,500                        |
| 1.3.3       | Static fire                  | 148,000          | Includes surge hire \$22,400                                   |
| 1.3.4       | Post-fire / pre-kitted spare | 48,000           | Engineers + spare handling                                     |
| 1.4.1       | LRR and RSO                  | 38,000           | Review board; payload witnesses                                |
| 1.4.2       | Range exclusion / marine     | 58,000           | Patrol boats + NOTAM                                           |
| 1.4.3       | Launch and telemetry capture | 82,000           | Launch-day operations                                          |
| 1.4.4       | Decrypt and handover         | 24,000           | Five-day engineering handover                                  |
| 1.4.5       | Demobilisation and audit     | 66,000           | Site restore + audit                                           |
| <b>Base</b> |                              | <b>1,500,000</b> |                                                                |

*Notes.* Labour: MA000010 C10 ordinary \$29.45/h (ppc 1 July 2026) from Annual Wage Review 2026 (Fair Work Commission, 2026). Employment on-cost = super 12% + annual leave 7.69% + personal leave 3.85% + public holidays 4.23% + LSL 2.17% + 17.5% leave loading on annual leave + workers compensation (1.8% professional / 4.5% pad) + QLD payroll tax 4.75% ( $1.378 \times \text{professional}$ ,  $1.405 \times \text{pad}$ ). Charge-out then divides by utilisation and applies overhead, with no consultancy fee: engineer/PM  $\$29.45 \times 1.378 / 0.72 \times 2.66 = \$150$ ; technician  $\$29.45 \times 1.405 / 0.85 \times 1.95 = \$95$ ; specialist  $\$29.45 \times 1.378 / 0.60 \times 2.44 = \$165$ . Surge is C10 × 1.5 overtime × pad on-cost /  $0.85 \times 1.37 = \$100$ . Crane: 8 lift days and 16 weld/NDT/bolt-up days at the same wet \$4,200 (QLD 100 t crawler band about \$3,800–\$4,500/day), cross-checked against Rawlinsons (Rawlinsons, 2025). Weather dollars sit only in RES (4 extra wet days). 1.1.3: ASA 280 h @ \$150 = \$42,000; CASA CRIS airspace instrument \$28,000; GBRMPA permission \$18,000 (Australian Space Agency, 2025; Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019). LOX/LN2 \$68,500 is a bulk industrial-gas analogue inside 1.3.2. Surge \$22,400 =  $4 \times 7 \text{ d} \times 8 \text{ h} \times \$100$  and sits in 1.3.3, not in contingency.

window, dual-path recorders, pre-kitted spare), so the pool uses residual  $P'_{\%} \times I_{\$}$ , not inherent  $P_{\%}$ . Summing the ten cost-bearing risks R-01, R-02, R-03, R-04, R-05, R-07, R-09, R-10, R-13 and R-14 gives \$35,760. The residual-uncertainty (RES) allowance is itemised in the same table and sums to \$43,130. Vehicle freight is out of scope and is not in RES. The two components sum to \$78,890, which is 5.26% of \$1,500,000, and the authorised  $BAC$  is therefore \$1,578,890.

Table 4.2: Contingency build-up. The 5.26% ratio is residual EMV plus four named residual items, not an input markup.

| Component                                                                   | Amount (AUD)     | Rule                                                              |
|-----------------------------------------------------------------------------|------------------|-------------------------------------------------------------------|
| $\Sigma$ residual EMV of ten cost-bearing risks                             | 35,760           | Residual $P'_{\%} \times I_{\$}$ ; treatments already in the base |
| LOX/LN2 price variance (18% of \$68,500 gases)                              | 12,330           | In-scope commodity; not freight                                   |
| Crane weather (only weather \$): 4 d $\times$ \$4,200                       | 16,800           | Extra wet days; R-06 is schedule-only                             |
| GN2/N2 purge: 40 G-size cylinders $\times$ \$200                            | 8,000            | Named multiply-out                                                |
| Cryogenic kit: 3 $\times$ 3 m LOX hoses $\times$ \$1,400 + seal kit \$1,800 | 6,000            | Named multiply-out                                                |
| <b>RES subtotal</b>                                                         | <b>43,130</b>    | Itemised; not a plug to 15%                                       |
| <b>Contingency reserve</b>                                                  | <b>78,890</b>    | 5.26% of base                                                     |
| <b><math>BAC</math></b>                                                     | <b>1,578,890</b> | Base + contingency                                                |

Several register rows are kept out of the \$35,760 so that money is not counted twice. R-08 (pad-civil defect) is contract-transferred under the design-and-construct package and carries no Gilmour EMV. R-06, R-11, R-15, R-16, R-17 and R-18 are schedule or HSE exposures without a residual dollar draw. R-12 is the TestFlight 1 in-flight failure mode; the vehicle is factory product, so site-ops EMV is \$0. If R-14 is realised and Option C is invoked, the \$21,000 is drawn from the \$78,890 reserve. It is not added to the \$1,500,000 work baseline. There is no management reserve above  $BAC$ : unknown unknowns would require a board variation this plan does not authorise.

#### 4.4 Time-phased budget and cash flow

The performance measurement baseline is the work estimate spread across the same dates as Table 3.1. Table 4.3 and Figure 4.1 use a uniform daily burn on each activity. October is the peak month because stacking, Wet Dress Rehearsal and the five-second static fire all sit there. December is \$6,000 because closeout (A-145) runs 21 November to 1 December. Only 1 December sits in that month, so the period is one day of demobilisation:  $\$66,000/11 = \$6,000$ . That is planned closeout, not unused contingency. Contingency is not monthly PV.

Table 4.3: Time-phased planned value of the \$1,500,000 work baseline. Contingency is not monthly spend.

| Month                                    | Period PV (AUD) | Cumulative PV (AUD) |
|------------------------------------------|-----------------|---------------------|
| Aug 2026                                 | 155,951         | 155,951             |
| Sep 2026                                 | 407,188         | 563,139             |
| Oct 2026 (peak: stack, WDR, static fire) | 628,861         | 1,192,000           |
| Nov 2026                                 | 302,000         | 1,494,000           |
| Dec 2026                                 | 6,000           | <b>1,500,000</b>    |

The S-curve therefore terminates at \$1,500,000. Figure 4.1 draws  $BAC = \$1,578,890$  as a horizontal authorisation line. Planned value at M-4 (15 October) is \$865,578—the same total used in Section 7.5. Planned value at Gate 1 (2 October) is \$602,301. Option C's \$21,000 is excluded from every period in Table 4.3. If it is spent, it appears as actual cost charged to contingency, not as a rewrite of the S-curve.

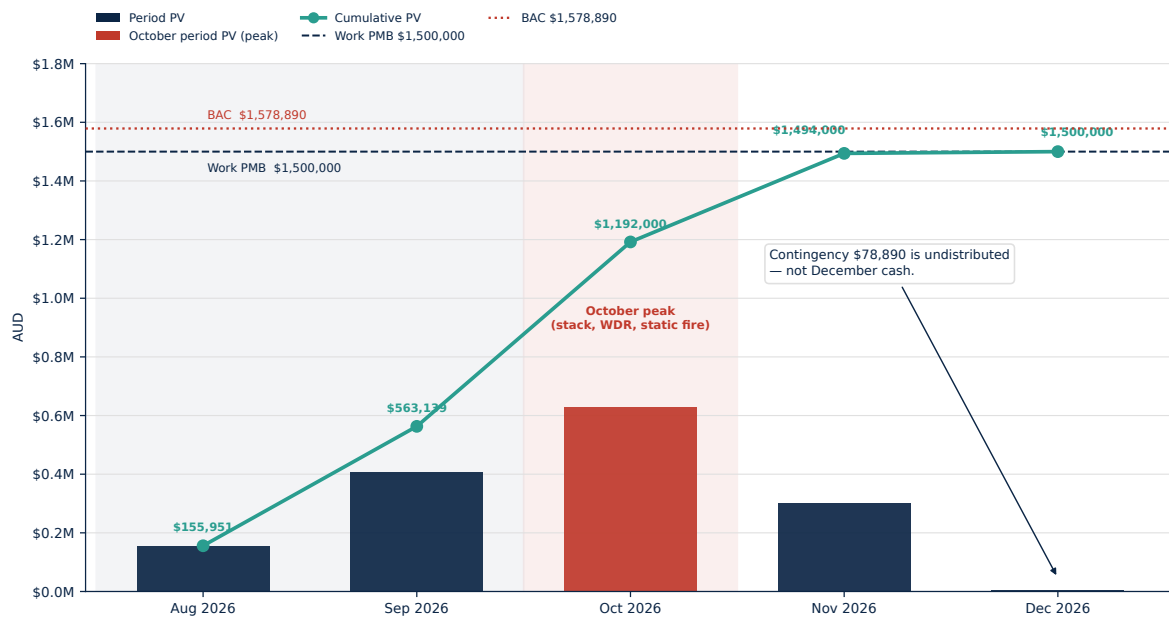


Figure 4.1: Cumulative planned-value S-curve for the Bowen campaign. Period bars are monthly PV; the teal line is cumulative PV and ends at the \$1,500,000 work baseline. The crimson dotted line is  $BAC$  (\$1,578,890) as an authorisation ceiling, not December cash. Contingency remains undistributed. Option C is not in the curve.

## 5 Resource management

Pad occupancy is a health-and-safety limit, not a roster convenience. The organic Bowen crew is eight named standing heads: GSE1–4, STK1–2, Launch Director and RSO. Juru, avionics and RF specialists sit against the blast-zone, suit-air and access-control cap of twelve people. Eight

is therefore the standing organisation; twelve is this PEP's occupancy ceiling that Figure 5.1 must never cross. Those occupancy and HSE numbers are pad controls in this PEP, not A1 charter figures. A1 finish-line HSE remains zero lost-time injuries, zero GBRMPA breaches and zero Juru heritage breaches.

## 5.1 Roles and responsibilities

Accountability is split across two layers, with exactly one Accountable owner per Level-3 package in Table 5.1. The consultancy layer is Abdul Wahab Obeid, Hadi Alkhub, Ziyad Alahmari and Ilsa Hashmi. The field layer is the Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager. Consulted names must see the work product before the Accountable owner signs; they are not a second A.

The pattern matches the network. Hadi is Accountable for stacking because the dawn-lift window is a schedule control (R-06). Hot operations are Accountable to the Launch Director with the RSO consulted, except range exclusion, which the RSO owns. Closeout audit is Ziyad's, so quality is not marking its own homework.

Table 5.1: RACI for Level-3 packages. Exactly one Accountable owner per row. Field titles are as used on the pad; consultancy names are the Group 1 officers.

| WBS   | Package                      | A               | R               | C                                           |
|-------|------------------------------|-----------------|-----------------|---------------------------------------------|
| 1.1.1 | PEP and integration          | Abdul           | Hadi            | Ziyad, Ilsa                                 |
| 1.1.2 | Governance and EVM           | Ilsa            | Hadi            | Abdul                                       |
| 1.1.3 | ASA / CASA / GBRMPA          | Abdul           | Hadi            | Launch Director                             |
| 1.1.4 | Juru heritage                | Juru Officer    | Ilsa            | Abdul, Ziyad                                |
| 1.2.1 | Receiving / cleanroom        | GSE Lead        | D&C mgr         | Hadi, Ilsa                                  |
| 1.2.2 | Crane and stacking           | Hadi            | GSE Lead        | Launch Director, RSO                        |
| 1.2.3 | Cryo GSE                     | D&C mgr         | GSE Lead        | Ziyad                                       |
| 1.2.4 | RF calibration               | Avionics        | Hadi            | Launch Director                             |
| 1.3.1 | Avionics power-on            | Avionics        | GSE Lead        | Hadi                                        |
| 1.3.2 | WDR (cold)                   | GSE Lead        | Juru Officer    | Ziyad, LD, RSO                              |
| 1.3.3 | Static fire                  | Launch Director | GSE Lead        | RSO, Ziyad, Avionics                        |
| 1.3.4 | Post-fire / pre-kitted spare | Avionics        | GSE Lead        | Ziyad                                       |
| 1.4.1 | LRR and RSO                  | Launch Director | RSO             | Abdul, Ziyad, Juru Officer, payload sponsor |
| 1.4.2 | Range exclusion              | RSO             | Launch Director | Juru Officer                                |
| 1.4.3 | Launch (T-0)                 | Launch Director | RSO, Avionics   | Abdul                                       |
| 1.4.4 | Telemetry handover           | Avionics        | Hadi            | Ilsa                                        |
| 1.4.5 | Demob and audit              | Ziyad           | D&C mgr         | Ilsa, Abdul                                 |

## 5.2 Capacity planning

Figure 5.1 and Table 5.2 are the weekly simultaneous occupancy implied by Table 3.1. Each activity carries a named pad crew. Daily occupancy is the unique-head union of every crew whose bar is live that calendar day; the weekly bar is the maximum of those daily counts. The HSE cap of twelve is a simultaneous limit, not a weekly unique-head total. Two RF specialists belong only to A-125. Mitigated A-125 uses *ES–EF* (29 August to 15 September), which is why weeks 3–6 sit two people above the unmitigated series. Unmitigated A-125 uses the late bar *LS–LF* (7–24 October), so those two heads sit in weeks 9–11—on Wet Dress Rehearsal, not on static fire after 24 October. Four overtime heads are an unmitigated overlay on 16–30 October only. Together with late RF they take 16 October to fourteen: two above the cap. That is R-13 in numbers.

The baseline does two things, in that order. First, A-125 is placed at *ES–EF* using its 39 days of total float. The move costs \$0 and does not change 15 November, because A-125 is a feeder into A-133, not a launch-critical bar. Second, four surge technicians are hired for seven days, 25–31 October (static fire plus the first diagnostics day; not 24 October, the last WDR day), inside WBS 1.3.3 at \$100/h loaded:  $4 \times 7 \times 8 \times \$100 = \$22,400$ , already inside the \$1,500,000 work estimate (Fair Work Commission, 2026). Those four replace the unmitigated overtime heads; they are not stacked on top of them. Mitigated 25–30 October is therefore eight named pad heads plus four surge, on the cap and never above it. 24 October is WDR only. Stacking ends 15 October and WDR starts 16 October, so there is no same-day overlap to invent. Option C, if invoked, shortens A-134 only and does not redraw Figure 5.1.

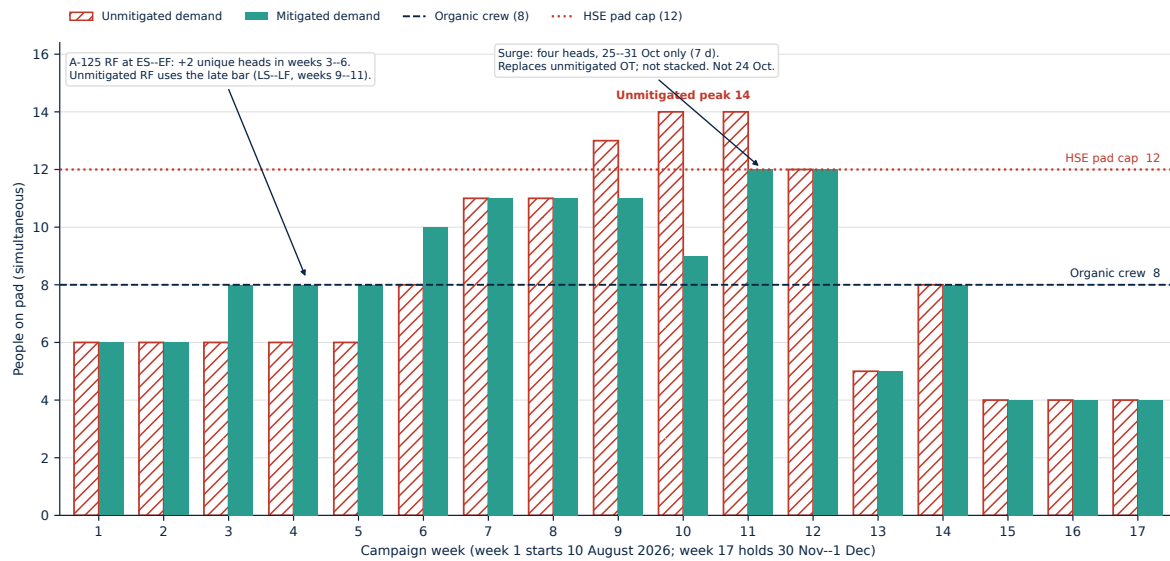


Figure 5.1: Weekly pad demand versus capacity, from Table 5.2. Each bar is the maximum simultaneous unique-head count that week. Hashed crimson bars are the unmitigated series (peak 14 on 16–24 October): Wet Dress Rehearsal plus A-125 on its late bar plus four overtime heads. Teal bars are the mitigated baseline (peak 12): RF at *ES–EF* in weeks 3–6, and four surge technicians on 25–31 October only. The navy dashed line is the organic crew of eight named heads; the crimson dotted line is this PEP’s HSE occupancy cap of twelve. Eight is not a hard pad limit.

Table 5.2: Weekly simultaneous pad occupancy, from activity  $\times$  named crew. Each weekly total is the maximum unique-head count on any calendar day that week (HSE cap is simultaneous, not weekly unique). Mitigated A-125 uses  $ES-EF$ ; unmitigated uses the late bar  $LS-LF$ . Surge is four heads on 25–31 Oct only (7 days). Unmitigated overtime is four heads on WDR and static fire (16–30 Oct). Base, RF, surge and OT are disjoint name sets, so they add on the peak day.

| Week | Mit. day | Base | RF | Surge | Mit. | Unmit. day | Base | RF | OT | Unmit. |
|------|----------|------|----|-------|------|------------|------|----|----|--------|
| 1    | 10 Aug   | 6    | 0  | 0     | 6    | 10 Aug     | 6    | 0  | 0  | 6      |
| 2    | 17 Aug   | 6    | 0  | 0     | 6    | 17 Aug     | 6    | 0  | 0  | 6      |
| 3    | 29 Aug   | 6    | 2  | 0     | 8    | 24 Aug     | 6    | 0  | 0  | 6      |
| 4    | 31 Aug   | 6    | 2  | 0     | 8    | 31 Aug     | 6    | 0  | 0  | 6      |
| 5    | 7 Sep    | 6    | 2  | 0     | 8    | 7 Sep      | 6    | 0  | 0  | 6      |
| 6    | 14 Sep   | 8    | 2  | 0     | 10   | 14 Sep     | 8    | 0  | 0  | 8      |
| 7    | 22 Sep   | 11   | 0  | 0     | 11   | 22 Sep     | 11   | 0  | 0  | 11     |
| 8    | 28 Sep   | 11   | 0  | 0     | 11   | 28 Sep     | 11   | 0  | 0  | 11     |
| 9    | 5 Oct    | 11   | 0  | 0     | 11   | 7 Oct      | 11   | 2  | 0  | 13     |
| 10   | 12 Oct   | 9    | 0  | 0     | 9    | 16 Oct     | 8    | 2  | 4  | 14     |
| 11   | 25 Oct   | 8    | 0  | 4     | 12   | 19 Oct     | 8    | 2  | 4  | 14     |
| 12   | 26 Oct   | 8    | 0  | 4     | 12   | 26 Oct     | 8    | 0  | 4  | 12     |
| 13   | 6 Nov    | 5    | 0  | 0     | 5    | 6 Nov      | 5    | 0  | 0  | 5      |
| 14   | 15 Nov   | 8    | 0  | 0     | 8    | 15 Nov     | 8    | 0  | 0  | 8      |
| 15   | 21 Nov   | 4    | 0  | 0     | 4    | 21 Nov     | 4    | 0  | 0  | 4      |
| 16   | 23 Nov   | 4    | 0  | 0     | 4    | 23 Nov     | 4    | 0  | 0  | 4      |
| 17   | 30 Nov   | 4    | 0  | 0     | 4    | 30 Nov     | 4    | 0  | 0  | 4      |

### 5.3 Conflict resolution and fallback

Resource conflicts escalate on the work, not on personalities. Tier 1 field leads have four hours to level, stagger or borrow from float without spending money. Tier 2 is the campaign PM: twelve hours to choose overtime, hire or resequence, with the cost consequence going to Ilisa so that Table 4.1 stays the book of record. Tier 3 is the Executive Board, with a 24-hour SLA, for any decision that consumes contingency or moves M-5 (30 October) or M-7 (15 November). Those gates are the same dates as Section 3; they are not a second calendar.

Team-member fallback is separate and is the A1 defect closed here. If a member is non-responsive for more than 48 hours, the Lead Planner audits the artefacts within 12 hours, the PM issues a WBS-reassignment addendum, and the Buddycheck PAF is locked against Teams and version history so that contribution evidence cannot be rewritten after the fact. That procedure reallocates packages in Table 5.1; it does not authorise a silent change to  $T-0$  or to  $BAC$ .

## 6 Risk management

Campaign risk is owned by Ziyad Alahmari (Risk and Quality Lead) and is scored on this pad, not as generic weather or budget-overrun rows. Eighteen cause–event–effect statements in Table 6.1 size the \$78,890 contingency already reported in Table 4.2, name the float that absorbs non-critical delay, and fix the owners who treat each item.

### 6.1 Risk approach and scales

The process follows ISO 31000 identify–analyse–evaluate–treat–monitor (International Organization for Standardization, 2018). Expected monetary value is calculated as follows: money is never read from the 1–5 heat-map scores. Inherent  $EMV = P_{\%} \times I_{\$}$  uses the dollar column in Table 6.1. Contingency uses residual  $EMV = P'_{\%} \times I_{\$}$  because the named treatments already sit in the \$1,500,000 base. Probability bands are  $P = 1$  below 5% (3% in EMV),  $P = 2$  from 5–20% (12%),  $P = 3$  from 21–40% (30%),  $P = 4$  from 41–60% (50%), and  $P = 5$  above 60% (70%). Schedule impact  $I_S$  is 1 (< 2 d), 2 (2–5 d), 3 (6–10 d), 4 (11–15 d) and 5 (> 15 d on a  $TF = 0$  path). Cost impact  $I_C$  uses a stated dollar scale: 1 (< \$15k), 2 (\$15k–\$39,999), 3 (\$40k–\$79,999), 4 (\$80k–\$149,999), 5 (\$150k or more). The matrix coordinate  $I$  is the maximum of those two bands. Residual  $P$  falls only where the named response actually bites. Residual  $I$  falls only where the effect itself is smaller; a 12–14 day recycle remains  $I = 4$  even if it is less likely. Ziyad re-scores the register at Gate 1 (2 October), at M-4 (15 October) and at M-5 (30 October), in the same cadence as the EVM dashboard in Section 7.5.

### 6.2 Integrated risk register

Table 6.1 is the full set. Cause, event and effect are three columns. Owners match the RACI in Table 5.1: Ziyad with the GSE Lead on static-fire abort (R-01) and township acoustics (R-17); the Launch Director on footprint, stage-separation and the in-flight TestFlight 1 mode (R-02, R-09, R-12); Ilsa Hashmi on LOX logistics, occupancy and the Juru commercial interface (R-10, R-13, R-04 with the Juru Cultural Officer); Hadi Alkhub on the dawn-lift window and countdown software (R-06, R-18); Abdul Wahab Obeid on permits and LRR deadlock (R-14, R-16). Figure 6.1 plots every inherent point and the residual after treatment. R-02, R-09 and R-12 share high-impact cells and are offset.

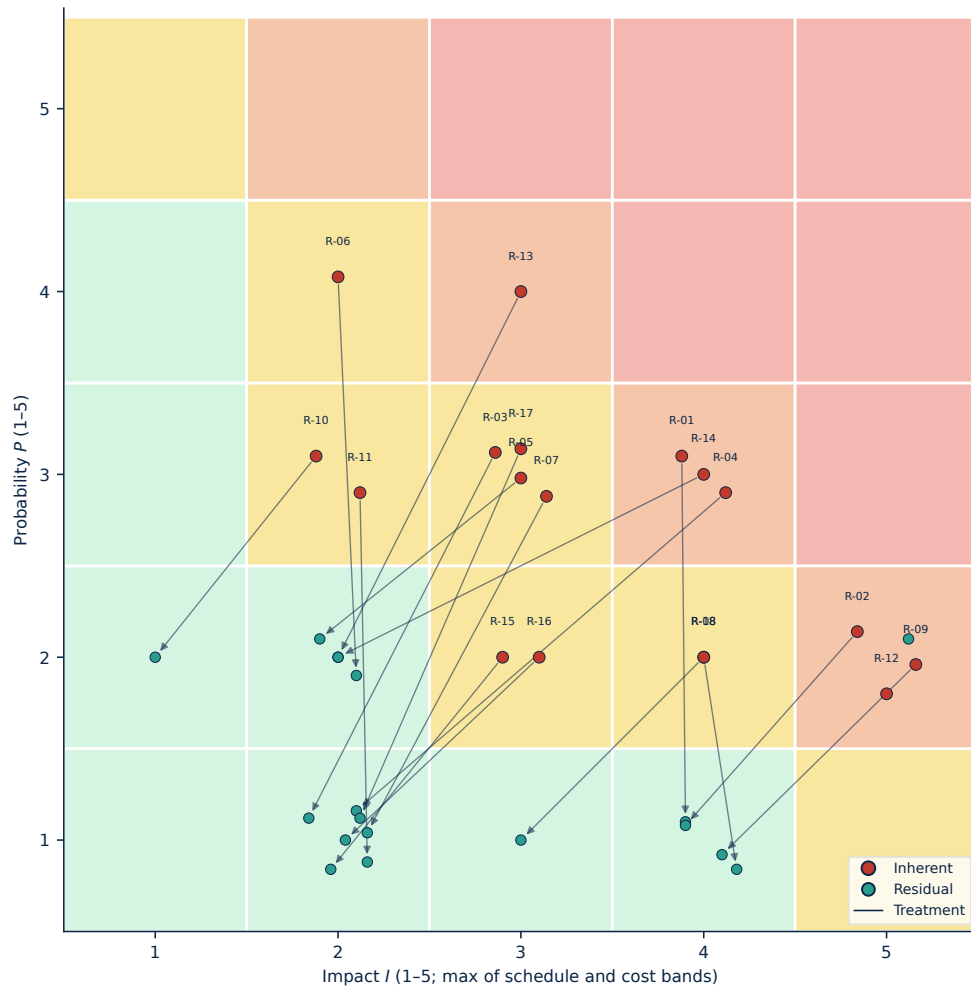


Figure 6.1: Inherent (crimson) to residual (teal) movement for all eighteen register items.  $P$  is vertical and  $I$  is horizontal. Most treatment arrows run down and left. R-01 keeps  $I = 4$  (a recycle is still 12–14 days). R-08 keeps  $I = 4$  (a hold-down failure is still a hot-ops stop; NDT drops  $P$  only). R-12 keeps  $I = 5$  (in-flight loss still ends heritage). R-02, R-09 and R-12 share high-impact cells and are offset.

Table 6.1: Integrated campaign risk register.  $P \times I$  uses the 1–5 scores. The EMV column is residual  $P'_{\%} \times I_{\$}$ , which feeds Table 4.2. A dash means the row is not in the \$35,760 pool.

| ID   | Anchor        | Cause                                         | Event                  | Effect                  | $P$ | $I$ | $P \times I$ | Owner                  | Response                                                                    | $P'$ | $I'$ | EMV   |
|------|---------------|-----------------------------------------------|------------------------|-------------------------|-----|-----|--------------|------------------------|-----------------------------------------------------------------------------|------|------|-------|
| R-01 | A-133 / 1.3.3 | Residual pump insulation                      | Static-fire auto-abort | 12–14 d recycle / \$90k | 3   | 4   | 12           | Ziyad / GSE Lead       | Thermal soak; pre-kitted LRU spare; 48 h swap. Design defects return to R&D | 1    | 4    | 2,700 |
| R-02 | A-143 / 1.4.3 | Trajectory energy toward park boundary        | Debris in GBRMPA zone  | Statutory halt          | 2   | 5   | 10           | Launch Director        | South-biased LRR footprint; joint GBRMPA protocol                           | 1    | 4    | 5,400 |
| R-03 | A-125 / A-143 | RF interference at Bowen array                | Telemetry <98.5%       | Heritage fail           | 3   | 3   | 9            | Avionics Lead          | Dual-path recorders; A-125 before hot fire; mobile backup                   | 1    | 2    | 1,500 |
| R-04 | A-114 / A-143 | Exclusion without stop-work                   | Juru complaint         | Range withheld          | 3   | 4   | 12           | Ilsa / Juru Officer    | Embedded monitors with stop-work; cultural HP before A-132                  | 1    | 2    | 1,200 |
| R-05 | A-124 / A-132 | Inadequate purge before LOX                   | Frozen/cracked valve   | WDR abort ~8 d          | 3   | 3   | 9            | GSE Lead               | LN2 pre-chill; spare valve kit; helium leak test HP-2                       | 2    | 2    | 6,600 |
| R-06 | A-123 / 1.2.2 | Abbot Point crosswind >20 kt at vertical mate | Lift abort             | Crane weather delay     | 4   | 2   | 8            | Hadi / Stack Conductor | Dawn-lift window. Weather \$ only in RES (4 extra wet days)                 | 2    | 2    | —     |

Table 6.1: Integrated campaign risk register (continued).

| ID   | Anchor        | Cause                            | Event                    | Effect         | <i>P</i> | <i>I</i> | <i>P</i> × <i>I</i> | Owner               | Response                                                                                          | <i>P'</i> | <i>I'</i> | EMV   |
|------|---------------|----------------------------------|--------------------------|----------------|----------|----------|---------------------|---------------------|---------------------------------------------------------------------------------------------------|-----------|-----------|-------|
| R-07 | A-124 / D&C   | D&C GSE IFC lags propulsion ICD  | Pad mismatch             | Rework         | 3        | 3        | 9                   | GSE Lead / PM       | ICD freeze gate; weekly integration; D&C LDs                                                      | 1         | 2         | 2,100 |
| R-08 | 1.2.3 / D&C   | Latent pad-civil defect          | Hold-down failure        | Hot-ops stop   | 2        | 4        | 8                   | D&C mgr (A: PM)     | NDT HP-1 drops <i>P</i> ; <i>I</i> stays 4 if a hold-down still fails; liability remains with D&C | 1         | 4         | 0*    |
| R-09 | A-143         | Stage-sep charge timing error    | Debris outside footprint | Inquiry / halt | 2        | 5        | 10                  | Launch Director     | Dual-string sep inhibit; wind hold toward park                                                    | 1         | 4         | 4,500 |
| R-10 | A-132 / 1.3.2 | Single 800 km LOX tanker delay   | WDR cannot load          | 2–5 d slip     | 3        | 2        | 6                   | Ilsa                | 72 h tanker confirm; second-tanker backup; RES on price                                           | 2         | 1         | 2,760 |
| R-11 | A-131         | Flight-computer vs pad EGSE skew | Power-on abort           | Avionics hold  | 3        | 2        | 6                   | Avionics Specialist | ICD freeze; cleanroom EGSE dry run before stack                                                   | 1         | 2         | —     |

Table 6.1: Integrated campaign risk register (continued).

| ID   | Anchor        | Cause                                  | Event                            | Effect                      | <i>P</i> | <i>I</i> | <i>P</i> × <i>I</i> | Owner                   | Response                                                                                   | <i>P'</i> | <i>I'</i> | EMV   |
|------|---------------|----------------------------------------|----------------------------------|-----------------------------|----------|----------|---------------------|-------------------------|--------------------------------------------------------------------------------------------|-----------|-----------|-------|
| R-12 | A-143         | Two-motor pump fault after <i>T</i> -0 | Thrust loss as at <i>T</i> +14 s | Heritage fail               | 2        | 5        | 10                  | Launch Director         | HP-3 screens ground abort; in-flight mode accepted ( <i>P</i> stays 2); no on-pad redesign | 2         | 5         | —     |
| R-13 | A-132 / A-133 | Peak 14 vs cap 12                      | Detank error                     | Recycle / injury            | 4        | 3        | 12                  | Ilisa                   | A-125 moved to Sep; four surge techs in base; 10 h stagger                                 | 2         | 2         | 3,600 |
| R-14 | A-113 / A-143 | Late ASA filing or CASA NOTAM          | 15 Nov slot refused              | Window missed               | 3        | 4        | 12                  | Abdul / Regulatory      | File 60 d prior (A1 Keep Satisfied); weekly ASA/CASA liaison; Option C backup              | 2         | 2         | 5,400 |
| R-15 | A-142         | Dugong in exclusion box                | Sweep extension                  | Delay inside 4 d A-142 hold | 2        | 3        | 6                   | RSO                     | Marine-mammal protocol; A-142 holds 4 d on the baseline and under Option C                 | 1         | 2         | —     |
| R-16 | A-141         | RSO vs Launch Director deadlock at LRR | Missed 10 Nov gate               | HP-4 unsigned               | 2        | 3        | 6                   | Abdul / Launch Director | Named DoA; escalate 24 h before A-141 <i>LF</i>                                            | 1         | 2         | —     |

Table 6.1: Integrated campaign risk register (continued).

| ID   | Anchor        | Cause                                    | Event                 | Effect              | $P$ | $I$ | $P \times I$ | Owner           | Response                                                             | $P'$ | $I'$ | EMV |
|------|---------------|------------------------------------------|-----------------------|---------------------|-----|-----|--------------|-----------------|----------------------------------------------------------------------|------|------|-----|
| R-17 | A-133         | Static-fire acoustic<br>>135 dB at Bowen | Complaint / hold      | Township stop       | 3   | 3   | 9            | Ziyad / HSE     | Water deluge;<br>1.5 km exclusion;<br>community notice<br>14 d prior | 1    | 2    | —   |
| R-18 | A-142 / A-143 | FTS/countdown abort at $T$ -10           | Recycle inside window | $T$ -0 day consumed | 2   | 4   | 8            | Hadi / Avionics | Two countdown rehearsals;<br>abort-to-recycle procedure              | 1    | 3    | —   |

\*R-08 is contract-transferred (EMV \$0 to Gilmour); residual  $I$  stays 4 because a hold-down that still occurs is still a hot-ops stop. R-06, R-11, R-12, R-15–R-18 are schedule, HSE or out-of-scope vehicle heritage; R-12 has no site-ops dollar draw. Residual EMV identities: R-01  $0.03 \times 90,000$ ; R-02  $0.03 \times 180,000$ ; R-03  $0.03 \times 50,000$ ; R-04  $0.03 \times 40,000$ ; R-05  $0.12 \times 55,000$ ; R-07  $0.03 \times 70,000$ ; R-09  $0.03 \times 150,000$ ; R-10  $0.12 \times 23,000$ ; R-13  $0.12 \times 30,000$ ; R-14  $0.12 \times 45,000$ . Sum \$35,760.

### 6.3 Top risks, contingency, float and delivery model

Three rows dominate board attention because they sit on the launch-critical path and can stop  $T-0$  or close the heritage case.

R-01 is a residual pump-insulation defect on A-133: auto-abort, a 12–14 day recycle, about \$90k of consumables. Ziyad and the GSE Lead treat it with thermal-soak instrumentation, a pre-kitted LRU spare and a 48-hour swap. Design defects still return to R&D. The score falls from (3, 4) to (1, 4): probability drops, but residual impact stays 4 because a recycle is still 12–14 days. Residual EMV is \$2,700 ( $3\% \times \$90k$ ), not the inherent \$27,000, because the soak, spare and swap already sit in the work baseline.

R-12 is the TestFlight 1 in-flight mode: thrust loss in two first-stage hybrid motors after  $T-0$ , as at  $T+14$  s, ending the heritage objective. The Launch Director owns HP-3 as the static-fire screen for a ground abort. HP-3 does not treat an event after  $T-0$ , so residual probability stays  $P' = 2$ . Residual impact stays 5 because an in-flight loss still closes heritage. There is no on-pad redesign and no site-ops dollar draw (EMV \$0).

R-14 is an ASA flight-authorisation or CASA NOTAM refusing the 15 November slot. Abdul files sixty days prior, as the charter scored ASA Keep Satisfied, and holds a weekly ASA/CASA liaison. Crashing A-123 does not treat R-14: permits already finish on 24 October with  $TF = 0$ . Option C is the schedule backup if this row fires. It is not justified by an invented 10–12 November payload geometry.

R-09 (stage-separation timing) and R-02 (footprint toward the park) remain high-impact register rows on the same LRR protocol. They are not the three board headlines.

Those ten cost-bearing rows sum to \$35,760 of residual EMV. Ilsa's itemised RES allowance of \$43,130 covers in-scope LOX price variance, crane weather (the only weather dollars), GN2/N2 purge stock and a cryogenic hose/seal kit. Together they are the \$78,890 in Table 4.2 (5.26% of base, not a 15% markup). Vehicle freight stays out of RES. R-06 is a dawn-lift schedule control with EMV \$0, so weather is not counted twice.

Float, not money, is the treatment for several feeders. A-125's 39 days let Ilsa park RF in weeks 3–6 so R-13 never meets an occupancy of 14. A-124's three days and A-114's five days absorb GSE and heritage delay before A-132 without moving 15 November. R-08 is transferred after HP-1 NDT; residual  $I$  stays 4. If R-14 fires, Abdul takes Option C to the Board and Ilsa draws \$21,000 from contingency. That draw is not in the \$1,500,000 work baseline, it does not move the M-7 diamond on Figure 3.2 until the Board says so, and A-142 remains a four-day marine hold.

## 7 Delivery and control

This section converts the baselines in Sections 3–6 into decision rights, a pad-specific procurement choice, hold points, Bowen HSE, and an earned-value framework that will be read at M-4.

It is a control design, not a progress report: the campaign has not been executed.

## 7.1 Governance and reporting

Governance is the organisation that will actually stop work, not a generic pyramid. The Gilmour Executive Board is the sponsor. Abdul Wahab Obeid is Campaign Project Manager and directs the three consultancy leads in Table 5.1. Field titles on Figure 7.1 match that table: Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager.

Delegation is numeric. Abdul may approve a variance of at most \$20,000 *and* at most 48 hours, and only if M-5 (30 October, A-133) and M-7 (15 November, A-143) do not move. Anything else—a contingency draw, Option C's \$21,000, a hold-point waiver, a *T*-0 change—goes to the Board within 24 hours. R-16 (RSO versus Launch Director deadlock) escalates 24 hours before A-141's late finish of 10 Nov, not inside the review room. The \$20,000 ceiling is below the dollar *impact* of the hot-fire rows that would recycle the pad (R-01 \$90,000; R-07 \$70,000). Residual EMVs on those rows are \$2,700 and \$2,100 because treatments already sit in the work baseline; a genuine recycle still cannot be buried in PM discretion.

Cadence matches the physical work. Field leads hold a daily pad standup. After Gate 1 (A-111, 2 Oct) Ilsa publishes a weekly EVM dashboard from A-112. Abdul takes a fortnightly pack to the sponsor. Daily standup does not rewrite Table 3.1. A-112 has 53 days of float so the control system does not compete with stacking for pad occupancy.

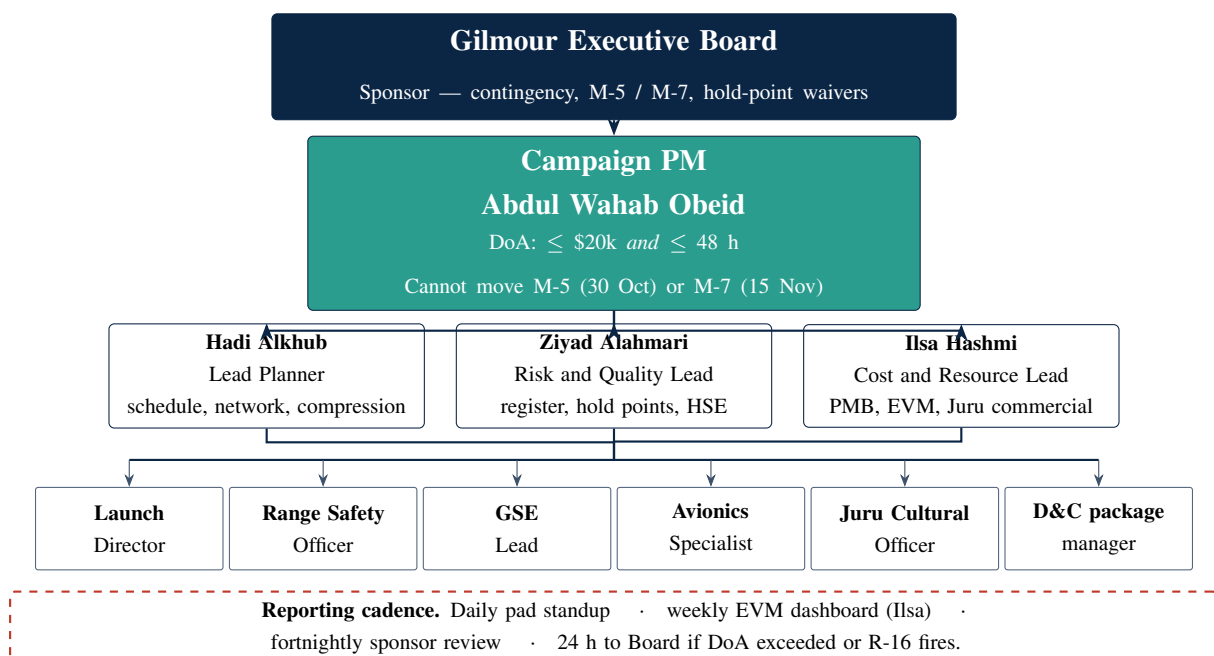


Figure 7.1: Campaign governance for Bowen pad operations. Named roles, a numeric DoA on the PM, and the reporting cadence. Field titles match Table 5.1. The Board, not the PM, moves M-5 or M-7 or draws the \$78,890 reserve.

## 7.2 Delivery model and procurement

The chosen delivery model is a managing-contractor / design-and-construct hybrid. Gilmour retains propulsion intellectual property, vehicle integration and the Australian launch licence. Civil works and cryogenic GSE packages are let as design-and-construct, with an interface-control freeze and liquidated damages. That split is the contractual answer to R-07 and R-08: an ICD lag is an LD event on the contractor, and a latent pad-civil defect stays with the D&C package after HP-1 NDT (R-08 EMV \$0 to Gilmour). Abdul stays Accountable on that row; the package manager stays Responsible.

Turnkey EPC was considered and rejected. After TestFlight 1, handing the propulsion stack and the launch authorisation to a single EPC would put flight-critical intellectual property and statutory accountability in a contractor who does not hold the licence (Gilmour Space Technologies, 2025; Australian Space Agency, 2025). Two tests decide the model. Scope certainty is high on site operations—seventeen Level-3 packages, locked dates, a \$1,500,000 work estimate—and low on the vehicle. Client capability is the other test: Gilmour can integrate a launch vehicle; it should not pour pad grout or run a 100 t crawler as a core competence. The charter supplied the 40/30/30 weights; A1 did not score the options. This PEP scores them for the site-ops choice only: Strategic Fit 40% (EPC 2/5, D&C 5/5), Technical Feasibility 30% (3/5 vs 4/5), Schedule 30% (3/5 vs 4/5), weighted 2.6 versus 4.4, so D&C wins. Alliance contracting was the second rejected alternative: a four-month campaign cannot carry pain-share overhead, and GBRMPA and CASA duties are not commercially shareable (Great Barrier Reef Marine Park Authority, 2019; Civil Aviation Safety Authority, 2024).

## 7.3 Quality

Quality is specified against AS9100D and ISO 9001:2015 (SAE International, 2016; International Organization for Standardization, 2015). Four hold points, each tied to a register row, are listed in Table 7.1. HP-1 is pad tie-down weld NDT on A-123 (R-08). HP-2 is the GSE helium leak test on A-124 (R-05). HP-3 is the pre-static-fire ignition interlock on A-133 (R-01). HP-4 is LRR sign-off on A-141: Launch Director, RSO and Juru Officer review with Abdul and Ziyad; the payload sponsor witnesses and does not sign. Work does not proceed past a hold without Ziyad's release or a Board waiver, and a Board waiver is outside Abdul's DoA.

Table 7.1: Quality hold points for the Bowen campaign. Each hold is a named artefact, not a meeting. Measurement rules in Section 7.5 follow the same IDs.

| Hold | Activity       | Risk | Artefact                                    | Release                                                 |
|------|----------------|------|---------------------------------------------|---------------------------------------------------------|
| HP-1 | A-123 (15 Oct) | R-08 | Pad tie-down weld<br>NDT                    | D&C mgr / Ziyad                                         |
| HP-2 | A-124 (12 Oct) | R-05 | GSE helium leak test                        | GSE Lead / Ziyad                                        |
| HP-3 | A-133 (30 Oct) | R-01 | Pre-static-fire ignition<br>interlock       | GSE Lead / Launch Director / Ziyad                      |
| HP-4 | A-141 (10 Nov) | R-16 | LRR sign-off (payload<br>sponsor witnesses) | Launch Director / RSO / Juru Officer /<br>Abdul / Ziyad |

Discrete tests and flight artefacts—A-132, A-133, A-141, A-143—are 0/100. A-113 is elapsed-day percent-complete because the work is a 76-day wait, not a binary test. Installations—A-123 and A-124—are milestone percent-complete tied to HP-1 and HP-2. LRR waits on A-134 rather than on A-133: HP-4 cannot be signed on incomplete post-fire data, which is why Option A was rejected (SAE International, 2016).

## 7.4 Health, safety and environment

HSE is this pad under the *Work Health and Safety Act 2011* (Qld) (Queensland Government, 2011). Liquid oxygen is the process hazard: LOX PPE, oxygen-clean tools, and exclusion during load and detank on A-132 and A-133. This PEP's pad controls—not A1 charter numbers—are a 1.5 km blast zone and twelve simultaneous heads, the same cap on Figure 5.1. Acoustic treatment is a water deluge so static-fire level at Bowen township stays at or below 135 dB (R-17), with community notice fourteen days prior. Juru artefact buffers and stop-work authority (R-04) are HSE controls: A-114 feeds A-132 as a cultural hold. Finish-line HSE remains as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches, with a five-second static fire of the four first-stage hybrid engines and handover of at least 98.5% of telemetry. Marine-mammal protocol on A-142 (R-15) is owned by the RSO (Great Barrier Reef Marine Park Authority, 2019). If R-14 fires, Option C shortens A-134 only and keeps A-141 at five days and A-142 at four days so HP-4 and the marine hold are not crashed. Occupancy never exceeds twelve on the mitigated histogram: R-13 is already treated in the \$1,500,000 base by moving A-125 and hiring four surge technicians.

## 7.5 Earned value management

Earned value is the instrument that will tell the Board whether the \$1,578,890 authorisation is still intact (Project Management Institute, 2019). It is a framework. The campaign has not started, so there is no fourteen-week EV/AC history to plot.

Distributed planned value is \$1,500,000, identical to Table 4.1 and to the S-curve in Figure 4.1. Contingency of \$78,890 is undistributed budget: it is not December cash and it is not monthly PV. Authorised  $BAC$  is \$1,578,890. Option C's \$21,000 is not in the work PMB; if R-14 fires and the Board invokes it, the draw appears as actual cost charged to contingency.

$SV$  and  $CV$  are currency, never days:

$$SV = EV - PV, \quad CV = EV - AC, \quad SPI = EV/PV, \quad CPI = EV/AC. \quad (5)$$

In-flight estimate at completion on the work baseline is

$$EAC = AC + (BAC_{\text{work}} - EV)/CPI, \quad (6)$$

with  $BAC_{\text{work}} = \$1,500,000$ . The same  $CPI$  is applied to the authorisation ceiling as  $BAC \times AC/EV$ , reported against \$1,578,890 (Project Management Institute, 2019). Finish-line rule: final  $AC \leq \$1,578,890$ . A Green  $CPI$  of 0.96 in October is allowed; finishing above  $BAC$  is not.

Thresholds that trigger action, used only in this section, are:

- **Green** —  $0.95 \leq CPI, SPI \leq 1.05$ : dashboard only.
- **Amber** —  $0.90 \leq \text{index} < 0.95$ : Exception Report and a seven-day recovery plan within 48 hours.
- **Red** —  $\text{index} < 0.90$ , or  $T-0$  slip  $> 2$  days: contingency freeze and Board intervention.  $SV$  is not written in days.

The only dashboard numbers permitted in this PEP are the two illustrative readings at M-4 (15 October 2026) in Table 7.2 and Figure 7.2. Planned value at that date is \$865,578. The planned label is HP-1 signed (A-123 100%), HP-2 signed, A-113 elapsed-day percent-complete. On-plan books  $EV$  equal to  $PV$  (\$865,578) and  $AC$  \$901,000, so  $CPI = 865,578/901,000 = 0.96$  and  $SPI = 865,578/865,578 = 1.00$ —Green. Even so,

$$EAC = 901,000 \times 1,500,000/865,578 = \$1,561,384,$$

which is identical to  $AC + (BAC_{\text{work}} - EV)/(EV/AC)$ . Authorisation

$$BAC \times AC/EV = 1,578,890 \times 901,000/865,578 = \$1,643,503$$

sits above the \$1,578,890 ceiling. Green at M-4 still carries a recovery obligation toward the finish-line rule.

The late-stack illustration holds  $PV$  at \$865,578 but drops  $EV$  to \$647,578 and  $AC$  to \$870,000 because HP-1 is unsigned. A-123 is milestone 0/100 tied to that hold, so the \$218,000

control account contributes \$0 of EV—the whole package is withheld, not a partial-hours claim. Then  $CPI = 647,578/870,000 = 0.74$  and  $SPI = 647,578/865,578 = 0.75$ —Red. Ilsa issues the Exception Report within 48 hours and freezes contingency. If R-14 has fired, the seven-day recovery invokes Option C and crashes A-134 only—not A-141 and not A-142. Work  $EAC$  is  $870,000 \times 1,500,000/647,578 = \$2,015,201$ .  $BAC \times AC/EV = 1,578,890 \times 870,000/647,578 = \$2,121,187$ . Those two points are a worked example at one physical gate, not a time series and not a claim that October has already been worked.

Table 7.2: Illustrative earned-value status at M-4 (15 October 2026). The campaign has not been executed. PV is the locked work-PMB value at stacking complete. Indices use the reported two-decimal CPI/SPI from  $EV/AC$  and  $EV/PV$ .

| Status at M-4                                 | PV      | EV      | AC      | CPI  | SPI  | Action                                             |
|-----------------------------------------------|---------|---------|---------|------|------|----------------------------------------------------|
| HP-1 signed (A-123<br>100Illustrative on-plan | 865,578 | 865,578 | 901,000 | 0.96 | 1.00 | Green — weekly dashboard                           |
| Illustrative late stack                       | 865,578 | 647,578 | 870,000 | 0.74 | 0.75 | Red — 48 h exception report;<br>contingency freeze |

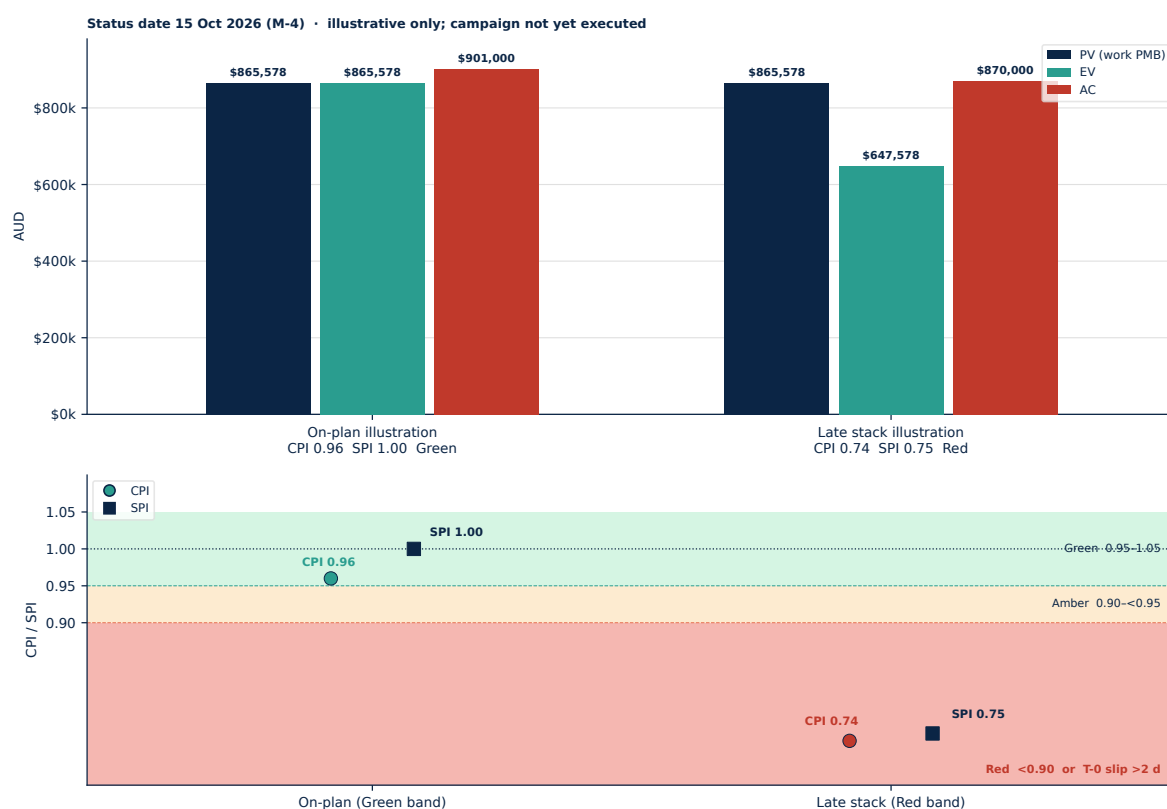


Figure 7.2: Illustrative status at M-4; campaign not yet executed. Upper panel: PV, EV and AC at the single status date 15 October 2026 for the on-plan and late-stack cases. Lower panel: Green / Amber / Red CPI–SPI bands, with the on-plan pair (0.96, 1.00) in Green and the late-stack pair (0.74, 0.75) in Red. Red is also triggered by  $T-0$  slip greater than 2 days. No weekly EV/AC history is implied.

## References

- SAE International. (2016). *Quality management systems—Requirements for aviation, space, and defense organizations* (AS9100D). SAE International.
- Rawlinsons. (2025). *Rawlinsons Australian construction handbook*. Rawlinsons Publishing.
- Fair Work Commission. (2026). *Annual Wage Review 2026* [2026] FWCFB 3500 (2 June 2026); *Manufacturing and Associated Industries and Occupations Award 2020* [MA000010] determination PR799291 (ppc 1 July 2026).
- Civil Aviation Safety Authority. (2024). *Cost recovery implementation statement: Aviation safety regulatory services*. Civil Aviation Safety Authority.
- International Organization for Standardization. (2018). *Risk management—Guidelines* (ISO 31000:2018). International Organization for Standardization.
- International Organization for Standardization. (2015). *Quality management systems—Requirements* (ISO 9001:2015). International Organization for Standardization.
- Project Management Institute. (2019). *The standard for earned value management*. Project Management Institute.
- Gilmour Space Technologies. (2025). *First test launch of Eris rocket a giant leap for Australian space capability*. Gilmour Space Technologies.
- Dougherty, R. (2026). *Gilmour confirms expectation to conduct second test flight in late 2026*. Space Connect.
- Smith, M. (2025). *Gilmour Space's Eris rocket fails initial test flight from Australia*. NASASpaceFlight.com.
- Australian Space Agency. (2025). *First launch attempt of Aussie made rocket*. Australian Government.
- Great Barrier Reef Marine Park Authority. (2019). *Permission system policy*. Great Barrier Reef Marine Park Authority.
- Work Health and Safety Act 2011* (Qld).